

PROBLEM STATEMENT NO 3- SUPERSTORE SALES DATASET

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Course: MBA (Finance & Business Analytics)

Mapped CO3: Prepare and transform the collected data into an analysis-ready dataset

1.Dataset Overview:

The Sample Superstore dataset from Kaggle contains retail sales information at the order level. It includes details about customers, products, sales, discounts, and profit for three main product categories: Furniture, Office Supplies, and Technology. The dataset also contains fields such as Order ID, Order Date, Ship Date, Ship Mode, customer information, location, category, and sub-category. In addition, Sales, Quantity, Discount, and Profit are recorded for each transaction.

Before using this information for charts or further analysis, the data needs to be checked and prepared properly. The sample used for this report contains 26 records. The preparation process focuses on finding errors, correcting the available information, and creating a clean dataset that can be used for further analysis.

2. Data Quality Issues Identified:

The first review of the 26 records showed a few issues that needed to be corrected before analysis:

- Duplicate record – One transaction was repeated with the same Order ID, Product ID, and other values.
- Missing values – One record did not have a value in the Discount column, while another record had a blank Profit value.
- Different date formats – The Order Date column contained dates written in different formats, such as 6/12/2016, 2016-06-12, and 06/09/2014.
- Different category names – The Category column had variations such as "furniture", "Furniture", "TECHNOLOGY", and "office supplies". If these values were not corrected, the same category could appear as separate categories during analysis.

3. Cleaning and Transformation Steps Applied:

3.1 Duplicate Removal

The records were checked for exact duplicates by comparing the complete row information. One repeated record was identified and removed. As a result, the sample was reduced from 26 records to 25 records.

3.2 Missing Value Handling

The blank value in the Discount column was replaced with 0, assuming that no discount was recorded for that transaction. The missing Profit value was estimated using a 30% margin of the Sales value. Since the actual cost information was not available, this estimated value was marked for review rather than treated as an exact figure.

3.3 Date Standardization

The Order Date and Ship Date columns were checked and converted into a common date format. All dates were standardized to YYYY-MM-DD. This makes it easier to sort the records and perform monthly, yearly, or other time-based analysis.

3.4 Category Value Correction

The Category column originally contained different versions of the same category names.

Before:

Furniture, Office Supplies, Technology, furniture, office supplies

After:

Furniture, Office Supplies, Technology

Extra spaces were removed and the category names were changed to a consistent format. This ensures that each product category is counted correctly in charts and summaries.

3.5 Calculated Columns

Two additional fields were created to support the analysis:

- Profit % – Calculated using $(\text{Profit} \div \text{Sales}) \times 100$. This helps compare profit margins between transactions with different sales amounts.

- Sales Category – Sales values were divided into four groups:
 - Low: ≤ 50
 - Medium: 50–200
 - High: 200–1000
 - Very High: > 1000

These groups make it easier to compare transactions in visualizations.

3.6 Grouping, Sorting, Filtering, and Aggregation

After cleaning the data, it was grouped by Category and Region. Total sales, total profit, and average profit margin were calculated for comparison. The results were arranged from the highest total sales to the lowest.

Transactions with a negative Profit % were also separated to identify records where the company made a loss.

4. Results on the Prepared Sample:

Sales and Profit by Product Category

Category	Total Sales (\$)	Total Profit (\$)	Orders
Furniture	4,822.52	217.19	5
Technology	2,032.06	175.08	2
Office Supplies	1,505.01	115.89	10

The table shows the total sales, profit, and number of orders for each product category in the prepared sample.

Sales and Average Margin by Region

Region	Total Sales (\$)	Avg. Profit %
West	5,481.79	23.52%
South	1,989.40	10.45%
Central	817.03	-55.20%
East	71.37	-1.43%

The regional summary shows that the West region recorded the highest sales in this sample. The Central region had a negative average profit percentage, which indicates that some transactions in that region resulted in losses.

After filtering the data for negative profit margins, 4 loss-making records were identified. Among them were a HEPA air filter with a -180% margin and a conference table sold with a 45% discount, which resulted in a -40% margin.

These examples show how the cleaned data can help identify transactions that require further investigation.

5. Final Prepared Dataset:

After completing the cleaning process, the dataset contains standardized dates, consistent category names, and no duplicate records. The missing Discount and Profit values were also addressed, and two additional fields—Profit % and Sales Category—were added.

The records were then arranged chronologically using Order Date. The resulting dataset is suitable for further work in Power BI, Tableau, or Excel PivotTables. The same approach can also be applied to the complete Superstore dataset, which contains approximately 9,994 rows.

The overall preparation process followed these steps:

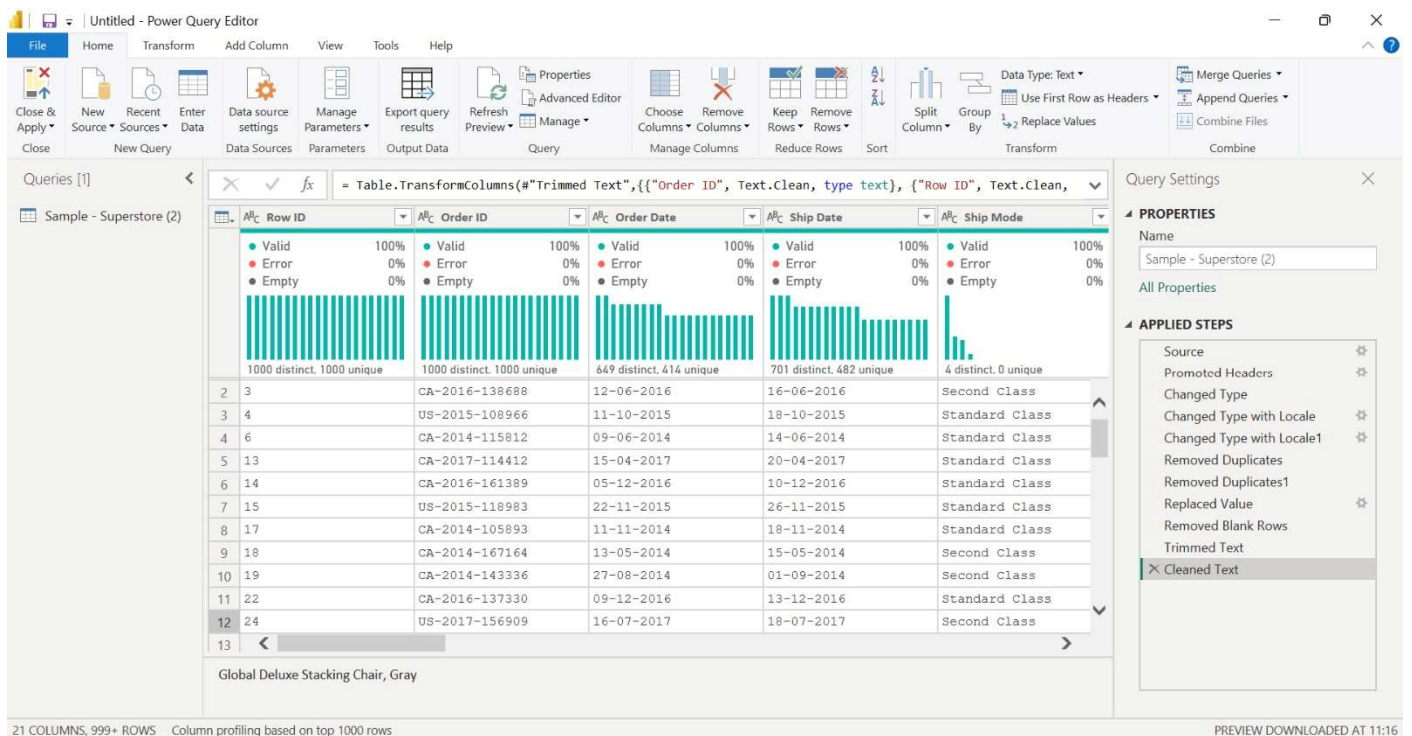
Remove duplicates → Handle missing values → Standardize dates → Correct category names → Create calculated fields → Group, sort, filter, and summarize the data.

6. Conclusion:

Cleaning the data is an important step before creating any dashboard or performing detailed analysis. In the Superstore sample, issues such as duplicate records, missing values, different date formats, and inconsistent category names could affect the accuracy of the results if they were left unchanged.

After making the required corrections, the data became easier to work with and provided a clearer view of sales and profit performance. The analysis also helped identify differences between product categories and regions, as well as

transactions that resulted in losses. This prepared dataset can therefore be used as a reliable starting point for creating dashboards and carrying out further analysis in Power BI, Tableau, or Excel.



DAX FUNCTION:

- PROFIT% = [Profit]/[Sales]*100
- ORDER_PROCESSING TIME = ('Sample - Superstore (2)'[Ship Date] - 'Sample - Superstore (2)'[Order Date])